RESOLUTION MSC.245(83)
(adopted on 12 October 2007)
RECOMMENDATION ON A STANDARD METHOD FOR
EVALUATING CROSS-FLOODING ARRANGEMENTSMSC 83/28/Add.2

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THE MARITIME SAFETY COMMITTEE,

RECALLING Article 28(b) of the Convention on the International Maritime Organization concerning the functions of the Committee,

RECALLING ALSO resolution A.266(VIII) entitled Recommendation on a standard method for establishing compliance with the re quirements for cross-flooding arrangements in passenger ships, adopted by the Assembly at its eighth session,

NOTING that the above Recommendation does not include provisions for cross-flooding arrangements other than pipes (i .e., cross-flooding times through ducts) or a provision to ensure adequate air ventilation for efficient cross-flooding (i.e., to account for the restrictive effect of air counter pressure during cross-flooding),

NOTING ALSO the revised SOLAS chapte r II-1 subdivision and damage stability

adequate air ventilation for efficient cross-flooding (i.e., to account for the restrictive effect of air counter pressure during cross-flooding),

NOTING ALSO the revised SOLAS chapte r II-1 subdivision and damage stability requirements for passenger and cargo ships, adopted by resolution MSC.216(82),

RECOGNIZING the need to establish a methodology for evaluating cross-flooding arrangements on ships subject to the applicable s ubdivision and damage stability requirements of SOLAS chapter II-1 to ensure uniform treatment of cross-flooding and equalization arrangements,

HAVING CONSIDERED the recommendations made by the Sub-Committee on Stability and Load Lines and on Fishing Vessels Safety at its fiftieth session,

1. ADOPTS the Recommendation on a standa rd method for evalua ting cross-flooding arrangements, the text of which is set out in the Annex to the present resolution;

2. INVITES Governments to apply the a nnexed Recommendation to cross-flooding calculations and to bring the Recommendation to the attention of all parties concerned.

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calculations and to bring the Recommendation to the attention of all parties concerned.

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1 Definitions

k : Sum of friction coefficients in the considered cross-flooding arrangement.

S ( )2m : Cross-section area of the cr oss-flooding pipe or duct. If the cross-section area is not circular, then:

4
2
equiv
equiv
D
S

=

where:
p
ADequiv = 4

A = actual cross-section area

p = actual cross-section perimeter

S ( )2m : Cross-section area of the cr oss-flooding pipe or duct. If the cross-section area is not circular, then:

4
2
equiv
equiv
D
S

=

where:
p
ADequiv = 4

A = actual cross-section area

p = actual cross-section perimeter

0 ()° : Angle before commencement of cros s-flooding. This assumes that the cross-flooding device is fully flooded but that no water has entered into the equalizing compartment on the opposite side of the damage (see appendix 1).

f ()° : Heel angle at final equilibrium (f).

()° : Any angle of heel between the commen cement of cross-flooding and the final equilibrium at a given time.

fW ( )3m : Volume of water which is used to bring the ship from commencement of cross-flooding 0 to final equilibrium f .

W ( )3m : Volume of water which is used to br ing the ship from any angle of heel to the final equilibrium f .

0H ()m : Head of water before commencement of cross-flooding, with the same assumption as for 0 .

H ()m : Head of water when any angle of heel is achieved.

fh ()m : Final head of water after cross-flooding (0=fh , when the level inside the

as for 0 .

H ()m : Head of water when any angle of heel is achieved.

fh ()m : Final head of water after cross-flooding (0=fh , when the level inside the equalizing compartment is equal to the free level of the sea).

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2 Formulae

2.1 Time required from comm encement of cross-flooding 0 to the final equilibrium f :

=
0
0
0
1
1
2
1
2
H
hgH
H
h
F S
W
T
f
f
f
f

2.2 Time required to bring the ship from any angle of heel to the final equilibrium f :

=

H
hgH
H
h
F S
WT
f
f
1
1
2
1
2

2.3 Time required from co mmencement of cross-flooding 0 until any angle of heel is achieved:

T T Tf =

2.4 Dimensionless factor of reduction of speed through an equaliza tion device, being a function of bends, valves, etc., in the cross-flooding system:

=
k
F 1

where F is not to be taken as more than 1.

Values for k can be obtained from appendix 2 or other appropriate sources.

function of bends, valves, etc., in the cross-flooding system:

=
k
F 1

where F is not to be taken as more than 1.

Values for k can be obtained from appendix 2 or other appropriate sources.

2.5 Cross-flooding through successive devi ces of different cross-section:

If the same flow crosses successive flooding devices of cross-section S 1, S 2, S 3 having corresponding friction coefficients k1, k2, k3, then the total k coefficient referred to S1 is:

$$k = k1 + k2 \cdot S12 / S22 + k3 \cdot S12 / S32$$

2.6 If different flooding devices are not cro ssed by the same volume, each k coefficient should be multiplied by the square of the ratio of the volume crossing the device and the volume crossing the reference section (which will be used for the time calculation):

$$k = k1 + k2 \cdot S12 / S22 \cdot W22 / W12 + k3 \cdot S12 / S32 \cdot W32 / W12$$

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2.7 For cross-flooding through devices in parallel that lead to the sa me space, equalisation

2.7 For cross-flooding through devices in parallel that lead to the same space, equalisation time should be calculated assuming that:

$S \cdot F = S1 \cdot F1 + S2 \cdot F2 +$

with $F = 1/k$ for each device of cross-section Si

3 Air pipe venting criteria

3.1 In arrangements where the total air pipe sectional area is 10% or more of the cross-flooding sectional area, the restrictive effect of any air back pressure may be neglected in the cross-flooding calculations. The air pipe sectional area should be taken as the minimum or the net sectional area of any automatic closing devices, if that is less.

3.2 In arrangements where the total air pipe sectional area is less than 10% of the cross-flooding sectional area, the restrictive effect of air back pressure should be considered in the cross-flooding calculations. The following method may be used for this purpose:

The k coefficient used in the calculation of cross-flooding time should take into account the drop

the cross-flooding calculations. The following method may be used for this purpose:

The k coefficient used in the calculation of cros s-flooding time should take into account the drop of head in the air pipe. This can be done using an equivalent coefficient ke, which is calculated according to the following formula:

$ke = kw + ka \cdot (a/w) \cdot (Sw/Sa)^2$

where:
kw = k coefficient for the cross-flooding arrangement (water)
ka = k coefficient for the air pipe
a = air density
w = water density
Sw = cross-section area of th e cross-flooding device (water)
Sa = cross-section of air pipe

4 Alternatives

As an alternative to the provisions in sections 2 and 3, and for arrangements other than those shown in appendix 2, direct cal culation using computational fluid dynamics (CFD), time-domain simulations or model testing may also be used.

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APPENDIX 1

EXAMPLE FOR TREATMENT OF HEEL ANGLES AND WATER HEADS
AT DIFFERENT STAGES OF CROSS-FLOODING

Figure 1

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APPENDIX 1

EXAMPLE FOR TREATMENT OF HEEL ANGLES AND WATER HEADS
AT DIFFERENT STAGES OF CROSS-FLOODING

Figure 1

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FRICTION COEFFICIENTS IN CROSS-FLOODING ARRANGEMENT

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FIGURE 1 FIG!;RE 3
I :'0 I ' I ' I I io 1,'. 1,', I . 10 .16 .ll
90· Ci RCULAR BEN D RADiUS BEND Rm · 2
FIGURE 4 fiGURE ~
, _____ cMo""e""ceO,',N""-_____ +-__ _ ____ ,~,~·CO~O~u~""O'~M~""~'~'C'~'~N~'~ _____ 1 r .. -.
PIPE INLET
k " 0.5
HG UR£ 8
\, .' V
~.-+--
" I
Th. valYe of k lCt!llUy ;nc:,uses wllh dc<:,e:ue in Froud.
numbt:. pUlic'Jluly bdo"'· 'peed. of 2 mI.tc.
:-40N.nETURN VALVE
C
V
,--+ ----· ----· ~4· _!_ '.
" c - "
It = 0.)
I--_____ -'00ATE VAl.VE
k - /0.0
DISC VA LVE
FICURE 11
FIG URE 9

numbt:. pUlic'Jluly bdo"'· 'peed. of 2 mI.tc.
:-40N.nETURN VALVE
C
V
,--+ ----· ----· ~4· _!_ '.
" c - "
It = 0.)
I--_____ -'00ATE VAl.VE
k - /0.0
DISC VA LVE
FICURE 11
FIG URE 9
The co cf!i ~i .nl above i. ~ mean yaJyC and dOcl in f~Cl vary
21 R.yn old'J nY,nlM:, (i.e. ~ ar;':, wilh V (0 : con ll1nl D .n d
~.) u ..... U n wjlh ,clu ivc ,,,.,&-'1,,c.1.
PIPE FRICTION LOSSES
fiGURE I I
k " O . ~
aUT HR F L Y VALVE. _____ __ _ MSC 83/28/Add.2
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Figure 13
Cross-flooding through a series of structural ducts with 1 manhole

1 00313 . 0 2748 . 0 < <+= ii L ifLk
4 17386 . 0 0212 . 1 6873 . 0 0986 . 023 + + = iiii L ifLLLk
434 . 1 >= iL ifk

Note: k is the friction coefficient related to each space between two adjacent girders. k is evaluated with effective cross-section area theref ore in calculations use the real cross-section area A and not S equiv. The pressure loss for entrance in the first manhole is already computed in the calculation, and k = 1 has to be added to take into account the outlet losses.

Li (in metres)
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in the calculation, and k = 1 has to be added to take into account the outlet losses.

Li (in metres)
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Figure 14
Cross-flooding through a series of structural ducts with 2 manholes

1 00627 . 0 4045 . 0 < <+= ii L ifLk
4 1356 . 0 1401 . 1 3593 . 0 0424 . 023 + = iiii L ifLLLk
417 . 1 >= iL ifk

Note: k is the friction coefficient related to each space between two adjacent girders. k is evaluated with effective cross-section area theref ore in calculations use the real cross-section area A and not S equiv. The pressure loss for entrance in the first manhole is already computed in the calculation, and k = 1 has to be added to take into account the outlet losses.

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APPENDIX 3

EXAMPLE USING FIGURES FOR A PASSENGER SHIP

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EXAMPLE USING FIGURES FOR A PASSENGER SHIP

Dimension of the considered cross-flooding pipe:

Diameter m D39 . 0=
Length m l0 . 21=
Cross-section area 212 . 0m S=
Wall thickness mm t5 . 17=

k-values for the considered cross-flooding system:

Inlet 0.45
Pipe friction D l02 . 0 1.08
2 radius bends (° =45) 0.36
Non-return valve 0.50
Outlet 1.00
= 39 . 3k

Sufficient air venting is assumed to be in place.

From this follows:

=
k
F 1

54 . 0
39 . 3
1 = =F

Time required from commencement of cross flooding o to the final equilibrium condition f :

=
0
0
0
1

1
2
1
2
H
hgH
H
h
F S
WT
f
f
f
f

Head of water before commencement of cross-flooding:

m H3 . 50 =

Volume of water which is used to bring the ship from comme ncement of cross-flooding to the final equilibrium condition:

3365m Wf =

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Volume of water which is used to bring the ship from comme ncement of cross-flooding to the final equilibrium condition:

3365m Wf =
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Final head of water after cross-flooding:

m hf 5 . 1=

min 0 . 12 721
3 . 5
5 . 11
1
3 . 5 81 . 9 2
3 . 5
5 . 11
54 . 0 12 . 0
365 2
2
2
3
= =

=
s T
m
mm
m
m
m
mT
f
s
mf

Time required to bring the vessel from the maximu m allowable angle of heel for final stage of flooding to the final equilibrium condition f :

=

H
hgH
H
h
F S
WT
f
f
1
1
2
1
*
2

Maximum allowable angle of heel for final stage of flooding ° =7

Head of water when the maximum allowable angle of heel
for final stage of flooding is achieved m H7 . 3=

Volume of water which is used to bring the vessel from the
maximum allowable angle of heel for final stage of flooding
to the final equilibrium condition 3160m W=

min 9 . 5 354
7 . 3
5 . 11
1
7 . 3 81 . 9 2
7 . 3
5 . 11

Volume of water which is used to bring the vessel from the maximum allowable angle of heel for final stage of flooding to the final equilibrium condition 3160m W=

min 9 . 5 354
7 . 3
5 . 11
1
7 . 3 81 . 9 2
7 . 3
5 . 11
54 . 0 12 . 0
160 2
2
2
3
= =

=
s T
m
mm
m
m
m
mT
s
m

Time required from comme ncement of cross-flooding o until the maximum allowable angle of heel for final stage of flooding is achieved:

T T Tf = = 12.0 min 5.9 min = 6.1 min

_____
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